- *Resource attributes (http://mng.bz/a1nY)*—Capture information about the entity for which telemetry data is recorded. They may capture an entire hierarchy of entity identification. You can use them to describe, for example, a Kubernetes node in a specific cluster in a specified cloud region or a specific container in a pod running an application you care about.
- Signal attributes are specific to a signal type, including
  - *Traces (http://mng.bz/gBMn)*—Semantic conventions, including for HTTP client and server spans, for remote procedure calls, such as gRPC (https://grpc.io/) spans, messaging system (Apache Kafka, RabbitMQ, etc.) spans, exceptions associated with a span, and many others.
  - *Metrics (http://mng.bz/e1XZ)*—Semantic conventions for HTTP client and server metrics; RPC client and server metrics; standard system metrics, like CPU usage or disk I/O; hardware; and a whole list of other metrics.
  - *Log (http://mng.bz/pP00)*—Attribute semantic conventions are, at the time of publication, still experimental in nature. Only two are defined: one for log media (describing the source of a log, such as the file name or I/O stream) and an event attribute that can, for example, describe the business context (e.g., browser or device).

What would a concrete example of a resource and signal attribute look like? Let us assume we are running an app that consists of certain services. Assume one of them is `svc-wow`, in a namespace called `boom`, in an EKS cluster called `prod-42`, in the `eu-west-1` region, and that service exposes HTTP metrics.

This scenario would be represented in OpenTelemetry with the pod as the signal source, so we would use these resource attributes:

- For the EKS cluster the service is located in, we would use the `cloud.region` (http://mng.bz/OxDO) attribute with a value of `eu-west-1` and the AWS-specific `aws.eks.cluster.arn` (http://mng.bz/Y12Q) attribute with a value such as `arn:aws:ecs:eu-west-1:123456789123:cluster/prod-42`.
- For the Kubernetes namespace the service is located in, we would use the generic resource attribute (http://mng.bz/GyQ8) `service.namespace` with a value of `boom` and `service.name` with a value of `svc-wow`.
- The signal type we're looking at here is metrics, which means we are using signal attributes to provide metadata on the metrics emitted. Let's say the `svc-wow` service exposes an HTTP API, and we want to report on the inbound request duration. We would use the `http.server.duration` (http://mng.bz/zXYg) metric with attributes such as `http.method`, `http.status_code`, and `net.host.name`. Coincidentally, the `http_status_code` attribute is an example of an attribute where not only the name is standardized but also its values—that is, the official list of HTTP response status codes (https://www.rfc-editor.org/rfc/rfc7231#section-6), as defined in RFC 7231.